Cold & Warm Episodes by Season

Oceanic Niño Index (ONI): Historical El Niño / La Niña episodes (1950–Present)

This page will continue to be updated by the 5th of each month, but per NWS Public Information Statement 26-05, the Relative Oceanic Niño index (RONI) is used for official monitoring and prediction of the El Niño-Southern Oscillation (ENSO) phenomenon.

Warm (red) and cold (blue) periods based on a threshold of +/- 0.5°C for the Oceanic Niño Index (ONI) [3 month running mean of ERSST.v5 SST anomalies in the Niño 3.4 region (5°N-5°S, 120°-170°W)], based on centered 30-year base periods updated every 5 years.

For historical purposes, periods of below and above normal SSTs are colored in blue and red when the threshold is met for a minimum of 5 consecutive overlapping seasons. The ONI is one measure of the El Niño-Southern Oscillation, and other indices can confirm whether features consistent with a coupled ocean-atmosphere phenomenon accompanied these periods.

Because of the high frequency filter applied to the ERSSTv5 data (Huang et al. 2017, J.Climate), ONI values may change up to two months after the initial "real time" value is posted. Therefore, the most recent ONI values should be considered an estimate.

Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ
1950	-1.5	-1.3	-1.2	-1.2	-1.1	-0.9	-0.5	-0.4	-0.4	-0.4	-0.6	-0.8
1951	-0.8	-0.5	-0.2	0.2	0.4	0.6	0.7	0.9	1.0	1.2	1.0	0.8
1952	0.5	0.4	0.3	0.3	0.2	0.0	-0.1	0.0	0.2	0.1	0.0	0.1
1953	0.4	0.6	0.6	0.7	0.8	0.8	0.7	0.7	0.8	0.8	0.8	0.8
1954	0.8	0.5	0.0	-0.4	-0.5	-0.5	-0.6	-0.8	-0.9	-0.8	-0.7	-0.7
1955	-0.7	-0.6	-0.7	-0.8	-0.8	-0.7	-0.7	-0.7	-1.1	-1.4	-1.7	-1.5
1956	-1.1	-0.8	-0.6	-0.5	-0.5	-0.5	-0.6	-0.6	-0.5	-0.4	-0.4	-0.4
1957	-0.2	0.1	0.4	0.7	0.9	1.1	1.3	1.3	1.3	1.4	1.5	1.7
1958	1.8	1.7	1.3	0.9	0.7	0.6	0.6	0.4	0.4	0.4	0.5	0.6
1959	0.6	0.6	0.5	0.3	0.2	-0.1	-0.2	-0.3	-0.1	0.0	0.0	0.0
Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ

1960	-0.1	-0.1	-0.1	0.0	0.0	0.0	0.1	0.2	0.3	0.2	0.1	0.1
1961	0.0	0.0	0.0	0.1	0.2	0.3	0.1	-0.1	-0.3	-0.3	-0.2	-0.2
1962	-0.2	-0.2	-0.2	-0.3	-0.3	-0.2	0.0	-0.1	-0.1	-0.2	-0.3	-0.4
1963	-0.4	-0.2	0.2	0.3	0.3	**0.5**	**0.9**	**1.1**	**1.2**	**1.3**	**1.4**	**1.3**
1964	**1.1**	**0.6**	0.1	-0.3	**-0.6**	**-0.6**	**-0.6**	**-0.7**	**-0.8**	**-0.8**	**-0.8**	**-0.8**
1965	**-0.6**	-0.3	-0.1	0.2	**0.5**	**0.8**	**1.2**	**1.5**	**1.9**	**2.0**	**2.0**	**1.7**
1966	**1.4**	**1.2**	**1.0**	**0.7**	0.4	0.2	0.2	0.1	-0.1	-0.1	-0.2	-0.3
1967	-0.4	-0.5	-0.5	-0.4	-0.2	0.0	0.0	-0.2	-0.3	-0.4	-0.3	-0.4
1968	-0.6	-0.7	-0.6	-0.4	0.0	0.3	0.6	0.5	0.4	**0.5**	**0.7**	**1.0**
1969	**1.1**	**1.1**	**0.9**	**0.8**	**0.6**	0.4	0.4	**0.5**	**0.8**	**0.9**	**0.8**	**0.6**
Year	**DJF**	**JFM**	**FMA**	**MAM**	**AMJ**	**MJJ**	**JJA**	**JAS**	**ASO**	**SON**	**OND**	**NDJ**
1970	**0.5**	0.3	0.3	0.2	0.0	-0.3	**-0.6**	**-0.8**	**-0.8**	**-0.7**	**-0.9**	**-1.1**
1971	**-1.4**	**-1.4**	**-1.1**	**-0.8**	**-0.7**	**-0.7**	**-0.8**	**-0.8**	**-0.8**	**-0.9**	**-1.0**	**-0.9**
1972	**-0.7**	-0.4	0.1	0.4	**0.7**	**0.9**	**1.1**	**1.4**	**1.6**	**1.8**	**2.1**	**2.1**
1973	**1.8**	**1.2**	**0.5**	-0.1	**-0.5**	**-0.9**	**-1.1**	**-1.3**	**-1.5**	**-1.7**	**-1.9**	**-2.0**
1974	**-1.8**	**-1.6**	**-1.2**	**-1.0**	**-0.9**	**-0.8**	**-0.5**	-0.4	-0.4	**-0.6**	**-0.8**	**-0.6**
1975	**-0.5**	**-0.6**	**-0.7**	**-0.7**	**-0.8**	**-1.0**	**-1.1**	**-1.2**	**-1.4**	**-1.4**	**-1.6**	**-1.7**
1976	**-1.6**	**-1.2**	**-0.7**	**-0.5**	-0.3	0.0	0.2	0.4	**0.6**	**0.8**	**0.9**	**0.8**
1977	**0.7**	**0.6**	0.3	0.2	0.2	0.3	0.4	0.4	**0.6**	**0.7**	**0.8**	**0.8**
1978	**0.7**	0.4	0.1	-0.2	-0.3	-0.3	-0.4	-0.4	-0.4	-0.3	-0.1	0.0
1979	0.0	0.1	0.2	0.3	0.2	0.0	0.0	0.2	0.3	**0.5**	**0.5**	**0.6**
Year	**DJF**	**JFM**	**FMA**	**MAM**	**AMJ**	**MJJ**	**JJA**	**JAS**	**ASO**	**SON**	**OND**	**NDJ**
1980	**0.6**	**0.5**	0.3	0.4	0.5	0.5	0.3	0.0	-0.1	0.0	0.1	0.0
1981	-0.3	-0.5	-0.5	-0.4	-0.3	-0.3	-0.3	-0.2	-0.2	-0.1	-0.2	-0.1
1982	0.0	0.1	0.2	**0.5**	**0.7**	**0.7**	**0.8**	**1.1**	**1.6**	**2.0**	**2.2**	**2.2**
1983	**2.2**	**1.9**	**1.5**	**1.3**	**1.1**	**0.7**	0.3	-0.1	**-0.5**	**-0.8**	**-1.0**	**-0.9**
1984	**-0.6**	-0.4	-0.3	-0.4	-0.5	-0.4	-0.3	-0.2	-0.2	**-0.6**	**-0.9**	**-1.1**
1985	**-1.0**	**-0.8**	**-0.8**	**-0.8**	**-0.8**	**-0.6**	**-0.5**	**-0.5**	-0.4	-0.3	-0.3	-0.4
1986	-0.5	-0.5	-0.3	-0.2	-0.1	0.0	0.2	0.4	**0.7**	**0.9**	**1.1**	**1.2**
1987	**1.2**	**1.2**	**1.1**	**0.9**	**1.0**	**1.2**	**1.5**	**1.7**	**1.6**	**1.5**	**1.3**	**1.1**
1988	**0.8**	**0.5**	0.1	-0.3	**-0.9**	**-1.3**	**-1.3**	**-1.1**	**-1.2**	**-1.5**	**-1.8**	**-1.8**
1989	**-1.7**	**-1.4**	**-1.1**	**-0.8**	**-0.6**	-0.4	-0.3	-0.3	-0.2	-0.2	-0.2	-0.1
Year	**DJF**	**JFM**	**FMA**	**MAM**	**AMJ**	**MJJ**	**JJA**	**JAS**	**ASO**	**SON**	**OND**	**NDJ**
1990	0.1	0.2	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.3	0.4	0.4
1991	0.4	0.3	0.2	0.3	**0.5**	**0.6**	**0.7**	**0.6**	**0.6**	**0.8**	**1.2**	**1.5**
1992	**1.7**	**1.6**	**1.5**	**1.3**	**1.1**	**0.7**	0.4	0.1	-0.1	-0.2	-0.3	-0.1
1993	0.1	0.3	0.5	0.7	0.7	0.6	0.3	0.3	0.2	0.1	0.0	0.1
1994	0.1	0.1	0.2	0.3	0.4	0.4	0.4	0.4	**0.6**	**0.7**	**1.0**	**1.1**
1995	**1.0**	**0.7**	**0.5**	0.3	0.1	0.0	-0.2	**-0.5**	**-0.8**	**-1.0**	**-1.0**	**-1.0**
1996	**-0.9**	**-0.8**	**-0.6**	-0.4	-0.3	-0.3	-0.3	-0.3	-0.4	-0.4	-0.4	-0.5
1997	-0.5	-0.4	-0.1	0.3	**0.8**	**1.2**	**1.6**	**1.9**	**2.1**	**2.3**	**2.4**	**2.4**
1998	**2.2**	**1.9**	**1.4**	**1.0**	**0.5**	-0.1	**-0.8**	**-1.1**	**-1.3**	**-1.4**	**-1.5**	**-1.6**
1999	**-1.5**	**-1.3**	**-1.1**	**-1.0**	**-1.0**	**-1.0**	**-1.1**	**-1.1**	**-1.2**	**-1.3**	**-1.5**	**-1.7**
Year	**DJF**	**JFM**	**FMA**	**MAM**	**AMJ**	**MJJ**	**JJA**	**JAS**	**ASO**	**SON**	**OND**	**NDJ**
2000	**-1.7**	**-1.4**	**-1.1**	**-0.8**	**-0.7**	**-0.6**	**-0.6**	**-0.5**	**-0.5**	**-0.6**	**-0.7**	**-0.7**
2001	**-0.7**	**-0.5**	-0.4	-0.3	-0.3	-0.1	-0.1	-0.1	-0.2	-0.3	-0.3	-0.3
2002	-0.1	0.0	0.1	0.2	0.4	**0.7**	**0.8**	**0.9**	**1.0**	**1.2**	**1.3**	**1.1**
2003	**0.9**	**0.6**	0.4	0.0	-0.3	-0.2	0.1	0.2	0.3	0.3	0.4	0.4
2004	0.4	0.3	0.2	0.2	0.2	0.3	**0.5**	**0.6**	**0.7**	**0.7**	**0.7**	**0.7**
2005	**0.6**	**0.6**	0.4	0.4	0.3	0.1	-0.1	-0.1	-0.1	-0.3	**-0.6**	**-0.8**
2006	**-0.9**	**-0.8**	**-0.6**	-0.4	-0.1	0.0	0.1	0.3	**0.5**	**0.8**	**0.9**	**0.9**
2007	**0.7**	0.2	-0.1	-0.3	-0.4	**-0.5**	**-0.6**	**-0.8**	**-1.1**	**-1.3**	**-1.5**	**-1.6**
2008	**-1.6**	**-1.5**	**-1.3**	**-1.0**	**-0.8**	**-0.6**	-0.4	-0.2	-0.2	-0.4	**-0.6**	**-0.7**
2009	**-0.8**	**-0.8**	**-0.6**	-0.3	0.0	0.3	**0.5**	**0.6**	**0.7**	**1.0**	**1.4**	**1.6**
Year	**DJF**	**JFM**	**FMA**	**MAM**	**AMJ**	**MJJ**	**JJA**	**JAS**	**ASO**	**SON**	**OND**	**NDJ**

2010	**1.5**	**1.2**	**0.8**	0.4	-0.2	**-0.7**	**-1.0**	**-1.3**	**-1.6**	**-1.6**	**-1.6**	**-1.5**
2011	**-1.3**	**-1.0**	**-0.8**	**-0.6**	**-0.5**	-0.4	-0.4	**-0.6**	**-0.8**	**-1.0**	**-1.0**	**-0.9**
2012	**-0.7**	**-0.6**	**-0.5**	-0.4	-0.2	0.1	0.3	0.4	0.4	0.3	0.1	-0.1
2013	-0.3	-0.3	-0.2	-0.2	-0.3	-0.3	-0.4	-0.3	-0.2	-0.1	-0.1	-0.2
2014	-0.3	-0.3	-0.1	0.2	0.3	0.2	0.1	0.1	0.3	**0.5**	**0.7**	**0.8**
2015	**0.7**	**0.6**	**0.7**	**0.8**	**1.0**	**1.3**	**1.6**	**1.9**	**2.2**	**2.5**	**2.6**	**2.8**
2016	**2.6**	**2.3**	**1.7**	**1.0**	**0.5**	0.0	-0.3	**-0.5**	**-0.6**	**-0.6**	**-0.6**	**-0.5**
2017	-0.2	0.0	0.2	0.3	0.4	0.4	0.2	-0.1	-0.3	**-0.6**	**-0.8**	**-0.9**
2018	**-0.8**	**-0.7**	**-0.6**	-0.4	-0.1	0.1	0.1	0.3	**0.5**	**0.8**	**1.0**	**0.9**
2019	**0.9**	**0.9**	**0.8**	**0.8**	**0.6**	**0.5**	0.3	0.2	0.2	0.4	**0.6**	**0.7**
Year	**DJF**	**JFM**	**FMA**	**MAM**	**AMJ**	**MJJ**	**JJA**	**JAS**	**ASO**	**SON**	**OND**	**NDJ**
2020	**0.6**	**0.6**	**0.5**	0.3	0.0	-0.2	-0.4	**-0.5**	**-0.8**	**-1.1**	**-1.2**	**-1.1**
2021	**-0.9**	**-0.8**	**-0.7**	**-0.5**	-0.4	-0.3	-0.3	-0.4	**-0.6**	**-0.8**	**-0.9**	**-0.9**
2022	**-0.8**	**-0.8**	**-0.9**	**-1.0**	**-0.9**	**-0.8**	**-0.8**	**-0.9**	**-1.0**	**-0.9**	**-0.8**	**-0.7**
2023	**-0.5**	-0.3	0.0	0.3	**0.6**	**0.8**	**1.1**	**1.4**	**1.6**	**1.8**	**2.0**	**2.1**
2024	**1.9**	**1.6**	**1.3**	**0.8**	**0.5**	0.2	0.1	-0.1	-0.2	-0.2	-0.3	-0.4
2025	-0.4	-0.2	-0.1	0.0	0.0	0.0	-0.1	-0.3	-0.4	-0.5	-0.6	-0.5
2026	-0.4	-0.1	0.1	0.5	1.0							

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